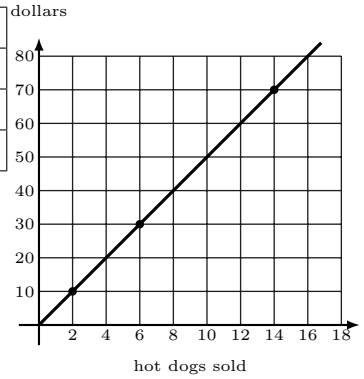
Artifact 12 – Calculating Slope and Rate of Change

Driving Question: From a table whose steps are not even, or from a graph, how do you find how much y changes for each 1 of x?

Example 1

A stadium cart clears the same amount on every hot dog.

Hot dogs	Dollars
2	10
6	30
14	70



Slope of the line:

Meaning in context:

Example 2

A dog walker charges the same amount for every dog.

Dogs	Dollars
2	15
5	37.50
9	67.50

Rate of change, dollars per dog. Show the two rows you used.

Would two other rows give the same rate? Show it.

Slope of the line

Notes to Forgetful Self

Rate of change, in my own words. From a table I

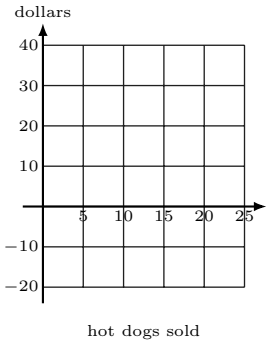
Artifact 13 – Slope and y-Intercept

Driving Question: Where does a line cross the y-axis, and what does that point mean in a real situation?

Example 1

The rent cart from your board: 12 dollars of rent, then 2 dollars on every hot dog. Fill the table, plot the points, draw the line.

Hot dogs	Dollars
0	
5	
10	
20	



Where does the line start? Write it as a point.

What does that point mean for the cart?

Rate of change, and the slope

Example 2

A student saves the same amount every week. The table starts at week 4; fill in the weeks above it.

Weeks	Dollars
0	
1	
2	
3	
4	90
8	150
15	255

Rate of change, dollars per week

Work back from week 4, one week at a time, to week 0.

The y-intercept, as a point, and what it means

Notes to Forgetful Self

The y-intercept, in my own words. On a graph it is